Date of Announcement: 28/10/22

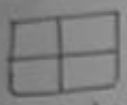
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Assignment-2

5. Explain the N-Queen's problem and its solution using Backtracking.(6 Marks)

→ N-Queens problem is to place n-queens in such a manner on an nxn chessboard that no queens attack each other by being in the same row, column or diagonal. For n=1, the problem has a trivial solution, and no solution exists for n=2 & n=3

N=2



No solution

N=3

Place 3 queens in 3x3 chess board

		q_1
q_2	$q_2 \times$	$q_2 \times$
$q_3 \times$	$q_3 \times$	$q_3 \times$

No solution.

→ So first we will consider the 4-queens problem and then generate it to n-queens problem. Given a 4x4 chessboard and number the rows and column of the chess board 1 through 4.

→ Since, we have to place 4 queens such as q_1 q_2 q_3 & q_4 on the chessboard, such that no two ~~queen~~ queens attack each other. Each queen must be placed on a different row ie, we put queen "i" on row "i"

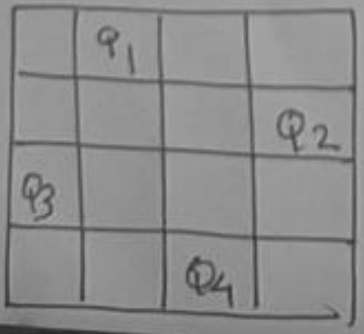
Solution

→ We place queen q_1 in the very first acceptable position(1,1)

→ Next, we put queen q_2 so that both these queens do not attack each other.

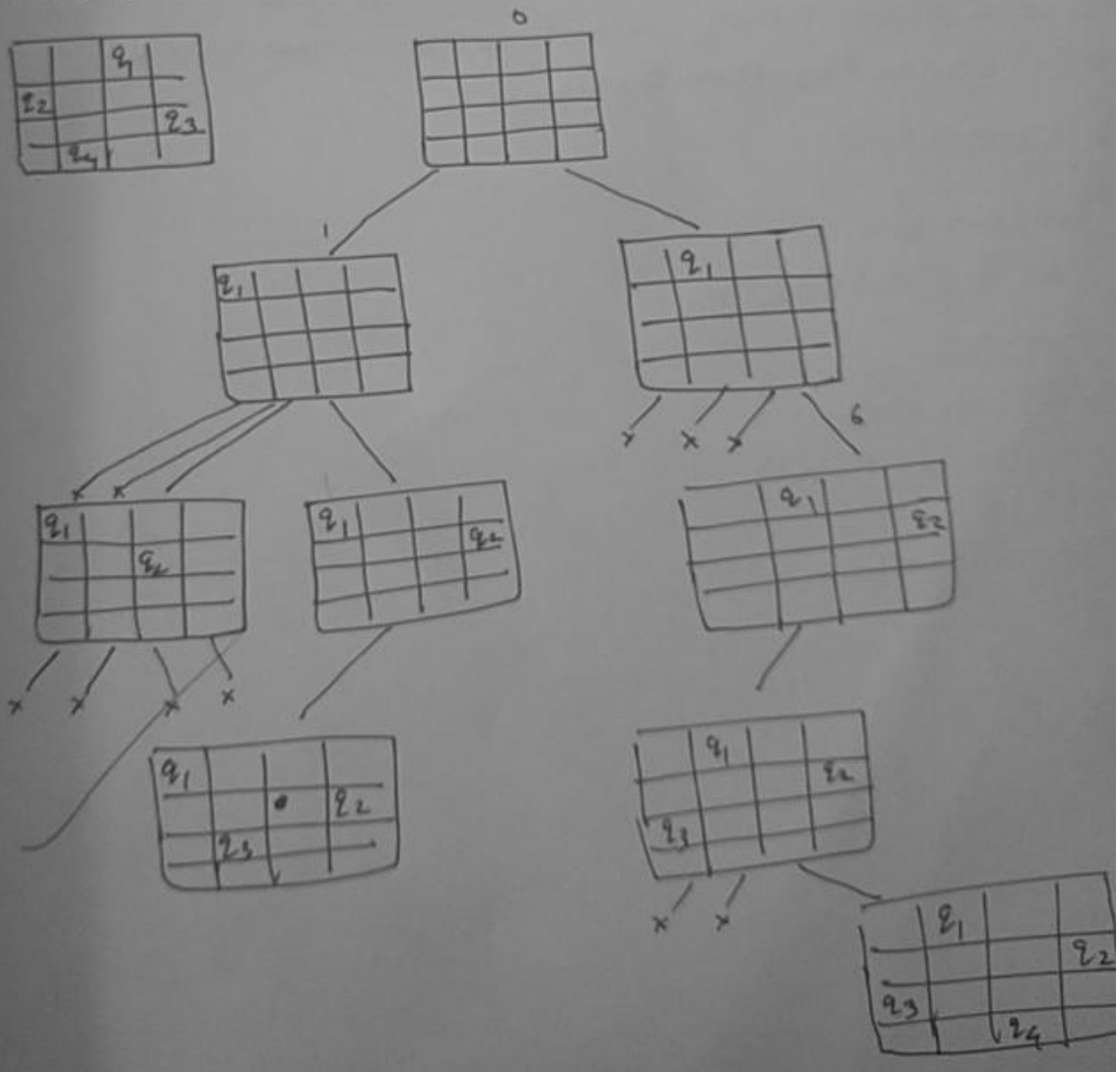
→ We find that if we place q_2 in column 1 & 2, then the dead end is encountered.

→ Thus the first acceptable position for q_2 in column 3, ie (2,3) but then no position is left for placing queen 'q^3' safely



is (3,2).

- But later this position also leads to a dead end, and no place is found where 'q_4' can be placed safely.
- Then we have to backtrack till 'q_3' and place it to (1,2) and then all other queens are placed safely by moving q_2 to (2,4) q_3 to (3,1) & q_4 to (4,3).
- That is we get the solution (2,4,1,3). This is one possible solution for the 4 queen problem.
- For another possible solution the whole method is repeated for all partial solutions.
- The other solutions for 4 queens problem is (3,1,4,



kth queen can
. It tests both whether i is
cosls $x_1 x_2 \ldots x_{k-1}$ and whether there is no other queen on
same diagonal.
. using place, we give a precise solution to the n-queens
problem.

```
place(k,i)
{
for j ← 1 to k-1
do if (x[j]=i)
or (Abs x([j] - i))=(Abs(j - k))
then return false;
return true;
}
```

. place (k,i) return true if a queen can be placed in the kth
row and i^{th} column otherwise return is false.
. x[] is a global array whose final k-1 values have been set
. Abs (r) returns the absolute value of r.

```
N-Queens(k,n)
{
for i ← 1 to n
do if place(k,i) then
{
x[k] ← i;
if(k==n) then
write(x[1...n]);
else
N-Queens(k+1,n);
}
}
```

3/11/2022